

Basic Electronics

Tutorial 2

- Referring to the circuit depicted in Fig. 1, count the number of (a) nodes; (b) elements; (c) branches.

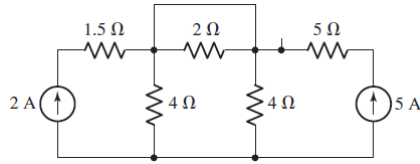


Figure 1: Prob.1

- Refer to the circuit of Fig. 2, and answer the following: (a) How many distinct nodes are contained in the circuit? (b) How many elements are contained in the circuit? (c) How many branches does the circuit have? (d) Determine if each of the following represents a path, a loop, both, or neither: (i) A to B (ii) B to D to C to E (iii) C to E to D to B to A to C (iv) C to D to B to A to C to E

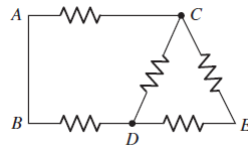


Figure 2: Prob.2

- In the circuit shown in Fig. 3, the resistor values are unknown, but the 2 V source is known to be supplying a current of 7 A to the rest of the circuit. Calculate the current labeled i_2 .

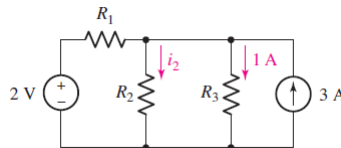


Figure 3: Prob.3

- In the circuit depicted in Fig. 4, i_x is determined to be 1.5 A, and the 9 V source supplies a current of 7.6 A (that is, a current of 7.6 A leaves the positive reference terminal of the 9 V source). Determine the value of resistor R_A .

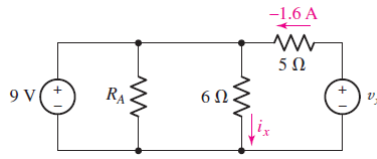


Figure 4: Prob.4

5. Determine the current labeled I_3 in the circuit of Fig. 5.

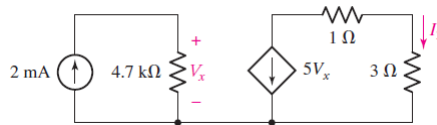


Figure 5: Prob.5

6. In the circuit of Fig. 6, it is determined that $v_1 = 3\text{ V}$ and $v_3 = 1.5\text{ V}$. Calculate v_R and v_2 .

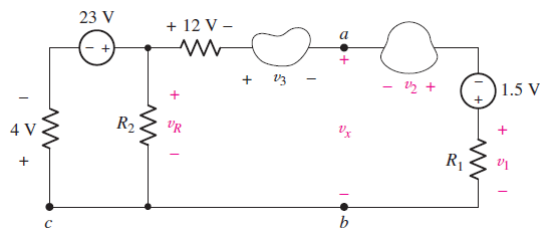


Figure 6: Prob.6

7. (a) Determine a numerical value for each current and voltage (i_1 , v_1 , etc.) in the circuit of Fig. 7. (b) Calculate the power absorbed by each element and verify that they sum to zero.
8. Compute the power absorbed by each element in the circuit of Fig. 8 if the mysterious element X is (a) a 13Ω resistor; (b) a dependent voltage source labeled $4v_1$, "+" reference on top; (c) a dependent voltage source labeled $4i_x$, "+" reference on top.

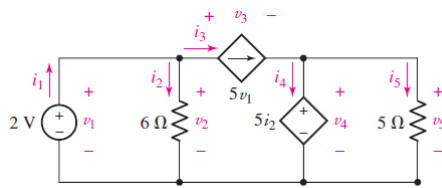


Figure 7: Prob.7

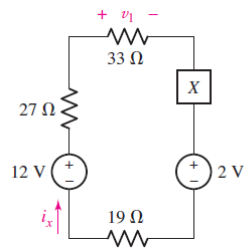


Figure 8: Prob.8

9. Referring to the circuit depicted in Fig. 9 , determine the value of the voltage v .

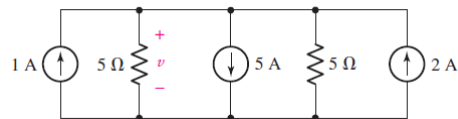


Figure 9: Prob.9

10. Employing resistance combination and current division as appropriate, determine values for i_1 , i_2 , and v_3 in the circuit of Fig. 10.

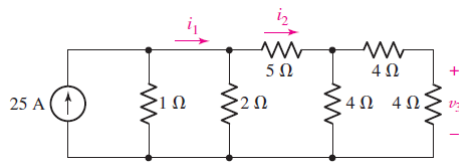


Figure 10: Prob.10

11. In the circuit of Fig. 11, only the voltage v_x is of interest. Simplify the circuit using appropriate resistor combinations and iteratively employ voltage division to determine v_x .

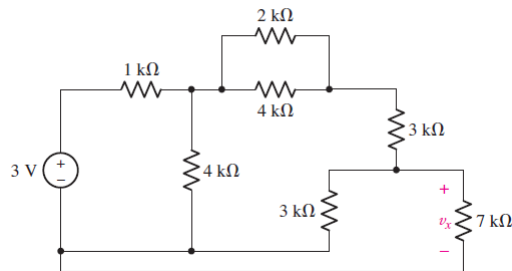


Figure 11: Prob.11